Course Learning Outcomes

- Describe positive and negative ways in which computer technology alters modes of social interaction at the personal level
- Evaluate ethical / social tradeoffs in technical decisions
- Evaluate the professional code of ethics from the ACM, the IEEE Computer Society, and other organizations
- Identify contemporary examples of intangible digital intellectual property
- Investigate the impact of technological solutions to privacy problems

Plagiarism

- What is it?
- Positive reasons for not plagiarizing
- Copying from single/multiple sources
- Paraphrasing, collusion
- How to avoid it?
- Reference list VS bibliography
- Plagiarism detection
- Penalty

ETHICS IN INFORMATION TECHNOLOGY

Chapter 1

An Overview of Ethics

What is Ethics?

- Moral code
  - Set of rules
  - Establishes boundaries of generally accepted behaviour
  - Different rules often have contradictions
- Morality
  - Social conventions about right and wrong
  - Widely shared
  - Form basis for an established consensus

What is Ethics? (cont'd.)

- Morality may vary by:
  - Age
  - Cultural group
  - Ethnic background
  - Religion
  - Life experiences
  - Education
  - Gender

Definition of Ethics

- Ethics
  - Set of beliefs about right and wrong behavior
- Virtues
  - Habits that incline people to do what is acceptable
- Vices
  - Habits of unacceptable behavior
- Virtues and vices define a personal value system
  - Scheme of moral values

The Importance of Integrity

- Integrity is a cornerstone of ethical behaviour
- People with integrity:
  - Act in accordance with a personal code of principles
  - Extend to all the same respect and consideration
  - Apply the same moral standards in all situations
- Lack of integrity emerges if you apply moral standards differently according to situation or people involved
- Many ethical dilemmas are not as simple as right versus wrong

The Difference Between Morals, Ethics & Laws

- Morals: one's personal beliefs about right and wrong
- Ethics: standards or codes of behavior expected of an individual by a group
- Law: system of rules that tells us what we can and cannot do
  - Laws are enforced by a set of institutions
  - Legal acts conform to the law
  - Moral acts conform to what an individual believes is the right belief of right and wrong

Ethics in the Business World

- Both the likelihood and the negative impact of inappropriate behavior have increased
- Several trends have increased the likelihood of unethical behavior:
  - Globalization creating complex work environments
  - Organizations challenged to maintain profits / revenue
  - Heightened vigilance by:
    - Employees
    - Shareholders
    - Regulatory agencies

Ethics in the Business World (cont'd.)

- Recent scandals in IT companies
  - Satyam Computer Services (India)
  - Hewlett Packard
  - Computer Associates International
  - IBM
- Not just executives, but even lower-level employees, can find themselves in the middle of an ethical dilemma

Why Fostering Good Business Ethics is Important

- To gain the good will of the community
- To create an organization that operates consistently
- To foster good business practices
- To protect organization/employees from legal action
- To avoid unfavorable publicity

Gaining the Good Will of the Community

- Organizations have fundamental responsibilities to society
  - Declared in formal statement of company's principles or beliefs
  - Include:
    - Making contributions to charitable organizations and non-profit institutions
    - Providing benefits for employees in excess of legal requirements
    - Choosing economic opportunities that might be more socially desirable than profitable

Gaining the Good Will of the Community (cont'd.)

- Socially responsible activities create good will
- Good will makes it easier for corporations to conduct business

Creating an Organization That Operates Consistently

- Consistency ensures that employees:
  - Know what is expected of them
  - Can employ the organization's values to help them in decision making
- Consistency also means that shareholders, customers, suppliers, and community know what they can expect of the organization

Fostering Good Business Practices

- Good ethics means good business/improved profits
- Companies that:
  - Produce safe and effective products
    - Avoid costly recalls and lawsuits
  - Provide excellent service that retains customers
  - Develop and maintain strong employee relations
    - Suffer lower turnover rates
    - Enjoy better employee morale

Fostering Good Business Practices (cont'd.)

- Suppliers/business partners place priority on working with companies that operate in a fair and ethical manner
- Bad ethics means bad business/waning profits
  - Bad ethics can lead to bad business results
  - Bad ethics can have a negative impact on employees

Protecting the Organization and Its Employees from Legal Actions

- U.S. Supreme Court established that an employer can be held responsible for the acts of its employees
- This principle is called *respondeat superior*
- Coalition of several legal organizations argues establishment of ethics and compliance programs should reduce criminal liability of organization
- Others argue company officers should not be given light sentences if their ethics programs are ineffective

Avoiding Unfavorable Publicity

- Public reputation of company strongly influences:
  - Value of its stock
  - How consumers regard products and services
  - Degree of oversight received from government
  - Amount of support and cooperation received
- Organizations are motivated to build strong ethics programs to avoid negative publicity

Improving Corporate Ethics

- Characteristics of a successful ethics program
  - Employees willing to seek advice about ethical issues
  - Employees feel prepared to handle situations that could lead to misconduct
  - Employees are rewarded for ethical behavior
  - Employees are not rewarded for success obtained through questionable means
  - Employees feel positive about their company

Appointing a Corporate Ethics Officer

- Corporate ethics officer
  - Provides vision and leadership in business conduct
  - Should be well-respected, senior-level manager who reports directly to the CEO
  - Ensures ethical procedures are put in place
  - Creates and maintains ethics culture
  - Is responsible for key knowledge/contact person for ethical issues

Ethical Standards Set by Board of Directors

- Board oversees the organization's business activities and management
- Board members of company are expected to:
  - Conduct themselves according to the highest standards of personal and professional integrity
  - Set standard for company-wide ethical conduct
  - Ensure compliance with laws and regulations
  - Create environment in which employees can seek advice about business conduct, raise issues, and report misconduct

Establishing a Corporate Code of Ethics

- Code of ethics
  - Highlights an organization's key ethical issues
  - Identifies overarching values and important principles
  - Focuses employees on areas of ethical risk
  - Offers guidance for employees to recognize and deal with ethical issues
  - Provides mechanisms to report unethical conduct
  - Help employees abide by the law, follow necessary regulations, and behave in an ethical manner

Establishing a Corporate Code of Ethics (cont'd.)

- Sarbanes-Oxley Act of 2002
  - Enacted in response to public outrage over several major accounting scandals
  - Section 404 requires that the CEO and CFO sign any SEC filing to attest to its accuracy
  - Section 406 requires public companies to disclose whether or not they have a code of ethics and if any waivers to that code have been granted

Establishing a Corporate Code of Ethics (cont'd.)

- Cannot gain company-wide acceptance unless it is:
  - Developed with employee participation
  - Fully endorsed by organization's leadership
- Must continually be applied to company's decision making and emphasized as part of its culture
- Breaches in the code of ethics must be identified and dealt with appropriately

Establishing a Corporate Code of Ethics (cont'd.)

1. Intel conducts business with honesty and integrity
2. Intel follows the letter and spirit of the law
3. Intel employees treat each other fairly
4. Intel employees act in the best interests of Intel and avoid conflicts of interest
5. Intel employees protect the company's assets and reputation

FIGURE 1-4 Intel's five principles of conduct

Credit: Five Principals of Conduct. © Intel Corporation. Reprinted by permission.

Conducting Social Audits

- Social audit
  - Reviews how well organization is meeting ethical and social responsibility goals
  - Communicates new goals for upcoming year
  - Shared broadly with employees, shareholders, investors, market analysts, customers, suppliers, government agencies, and local communities

Requiring Employees to Take Ethics Training

- Personal convictions improved through education
- Comprehensive ethics education program encourages employees to act responsibly and ethically
  - Often presented in small workshop formats
  - Employees apply code of ethics to hypothetical but realistic case studies
  - Demonstration of recent company decisions based on principles from the code of ethics

Requiring Employees to Take Ethics Training (cont'd.)

- Critical that training increase the percentage of employees who report incidents of misconduct
- Employees must:
  - Learn effective ways of reporting incidents
  - Be reassured their feedback will be acted on without retaliation

Including Ethical Criteria in Employee Appraisals

- Only 43% of companies include ethical conduct in employee's performance appraisal
- Ethical criteria include:
  - Treating others fairly and with respect
  - Operating effectively in a multicultural environment
  - Accepting personal accountability
  - Continually developing themselves and others
  - Operating openly and honestly with all

Creating an Ethical Work Environment

- Good employees may make bad ethical choices
- May be encouraged to do “whatever it takes” to get the job done
- Employees need a knowledgeable resource to discuss perceived unethical practices
  - A manager
  - Legal or Internal Audit Department
  - Business Unit’s legal counsel
  - Anonymously through internal Web site

Creating an Ethical Work Environment (cont'd.)

TABLE 1-3 Manager's checklist for establishing an ethical work environment

Question	Yes	No
Does your organization have a code of ethics?		
Do employees know how and to whom to report any infractions of the code of ethics?		
Do employees feel that they can report violations of the code of ethics safely and without fear of retaliation?		
Do employees feel that action will be taken against those who violate the code of ethics?		
Do senior managers set an example by communicating the code of ethics and using it in their own decision making?		
Do managers evaluate and provide feedback to employees on how they operate with respect to the values and principles in the code of ethics?		
Are employees aware of sanctions for breaching the code of ethics?		
Do employees use the code of ethics in their decision making?		

Source Line: Course Technology/Cengage Learning.

Including Ethical Considerations in Decision Making

- Steps in a decision-making process
  - Develop problem statement
  - Identify alternatives
  - Evaluate and choose alternative
  - Implement decision
  - Evaluate results
  - Success

Develop a Problem Statement

- Clear, concise description of the issue
- Answers these questions:
  - What causes people to think there is a problem?
  - Who is directly affected by the problem?
  - Is there anyone else affected?
  - How often does it occur?
  - What is the impact of the problem?
  - How serious is the problem?
- Most critical step in decision-making process

Develop a Problem Statement (cont'd.)

- Example of a good problem statement:
  - "Our product supply organization is continually running out of stock of finished products, creating an out-of-stock situation on over 15 percent of our customer orders, resulting in over $300,000 in lost sales per month."
- Examples of poor problem statements:
  - "We need to implement a new inventory control system." (possible solution, not a problem statement)
  - "We have a problem with finished product inventory." (not specific enough)

Identify, Evaluate, and Choose an Alternative

- Enlist help to brainstorm alternative solutions
- Evaluate by weighing laws, guidelines, and principles
- Consider likely consequences of each alternative
- Alternative selected must:
  - Be ethically and legally defensible
  - Be consistent with policies and code of ethics
  - Take into account impact on others
  - Provide a good solution to problem

Common Approaches to Ethical Decision Making

TABLE 1-4 Four common approaches to ethical decision making

Approach to dealing with moral issues	Principle
Virtue ethics approach	The ethical choice best reflects moral virtues in yourself and your community.
Utilitarian approach	The ethical choice produces the greatest excess of benefits over harm.
Fairness approach	The ethical choice treats everyone the same and shows no favoritism or discrimination.
Common good approach	The ethical choice advances the common good.

Source Line: Course Technology/Cengage Learning.

Virtue Ethics Approach

- Virtue ethics approach
  - Focuses on concern with daily life in a community
  - People guided by virtues to reach “right” decision
  - More effective than following set of principles/rules
- Problems
  - Does not provide guide for action
  - Virtue cannot be worked out objectively; depends on circumstances

Utilitarian Approach

- Utilitarian approach
  - Chooses action that has best overall consequences
  - Finds the greatest good by balancing all interests
  - Fits concept of value in economics and the use of cost-benefit analysis
- Problems
  - Measuring and comparing values is often difficult
  - Predicting resulting benefits and harm is difficult

Fairness Approach

- Fairness approach
  - Focuses on fair distribution of benefits/burdens
  - Guiding principle is to treat all people the same
- Problems
  - Decisions can be influenced by personal bias
  - Others may consider the decision unfair

Common Good Approach

- Common good approach
  - Work together for common set of values and goals
  - Implement systems that benefit all people
- Problems
  - Consensus is difficult
  - Some required to bear greater costs than others

Implement the Decision and Evaluate the Results

- Implement the decision
  - Efficient, effective, timely implementation
  - Communication is key for people to accept change
  - Transition plan made easy and pain-free
- Evaluate the results
  - Monitor results for desired effect
  - Observe impact on organization and stakeholders
  - Return to “Develop problem statement” step if further refinements may be needed

Ethics in Information Technology

- Public concern about the ethical use of information technology includes:
  - E-mail and Internet access monitoring
  - Downloading in violation of copyright laws
  - Unsolicited e-mail (spam)
  - Hackers and identify theft
  - Students and plagiarism
  - Cookies and spyware

Ethics in Information Technology (cont'd.)

- The general public does not understand the critical importance of ethics as applied to IT
- Important decisions are often left to technical experts
- General business managers must assume greater responsibility for these decisions by:
  - Making decisions based on technical savvy, business know-how, and a sense of ethics
  - Creating an environment where ethical dilemmas can be discussed openly, objectively, and constructively

Ethics in Information Technology (cont'd.)

- Goals of this text
  - To educate people about the tremendous impact of ethical issues in the successful and secure use of information technology
  - To motivate people to recognize these issues when making business decisions
  - To provide tools, approaches, and useful insights for making ethical decisions

Summary

- Ethics is important because the risks associated with inappropriate behavior have increased
- Organizations have at least five good reasons for encouraging employees to act ethically
  - To gain the good will of the community
  - To create an organization that operates consistently
  - To foster good business practices
  - To protect the organization and its employees against legal action
  - To avoid unfavorable publicity

Summary (cont'd.)

- Organizations require successful ethics programs
- The corporate ethics officer ensures that ethical procedures are installed and followed
- Managers' behaviors and expectations can strongly influence employees' ethical behavior
- Most of us have developed a simple decision-making model that includes five steps
- Ethical considerations must be incorporated into decision making

Summary (cont'd.)

- Four common approaches to ethical decision making
  - Virtue ethics approach
  - Utilitarian approach
  - Fairness approach
  - Common good approach